

MPM — Safety Specification v1.0

Microsoft-Aligned Standards Edition

Takeaway: The Safety Specification defines the **complete safety envelope** for the MPM: hazard zones, protective geometry, sensor requirements, interlock logic, emergency stop behavior, and compliance thresholds. It ensures that any MPM-compatible module can operate safely across manufacturers, sovereigns, and hyperscalers.

1. Safety Model Overview

The MPM safety model consists of five coordinated layers:

- **Hazard Envelope**
- **Protective Geometry**
- **Sensor & Interlock System**
- **Emergency Stop Behavior**
- **Fault Handling & Logging**

Each layer is mandatory for certification.

2. Hazard Envelope

The hazard envelope defines the spatial zones around the rotating mechanical interface.

2.1 Hazard Zones

- **Zone 0 — Contact Hazard Zone** Radius: **0–35 mm** from shaft axis Direct contact with rotating components. Must be fully enclosed.
- **Zone 1 — Proximity Hazard Zone** Radius: **35–65 mm** Risk of entanglement or impact. Must be shrouded.
- **Zone 2 — Safe Interaction Zone** Radius: **≥ 65 mm** No rotating components. User interaction permitted.

2.2 Envelope Requirements

- Zone 0 must be fully enclosed by the safety shroud.
- Zone 1 must be partially enclosed with no direct access.
- Zone 2 must remain unobstructed for controller access.
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3. Protective Geometry

Protective geometry ensures physical isolation of hazardous components.

3.1 Safety Shroud

- Radius: **65 mm** minimum
- Depth: **70 mm** minimum
- Material: Impact-resistant polymer or composite
- Flame rating: UL94-V0 or equivalent
- Clearance to rotating shaft: **≥ 3 mm**

3.2 Locking Mechanism Guard

- Must prevent finger access
- Minimum aperture: **≤ 6 mm**
- Must remain engaged during operation

3.3 Cable Routing Protection

- Bend radius: **≥ 15 mm**
- No cable may enter Zone 0 or Zone 1

3.4 Structural Frame Safety

- No sharp edges
- Minimum edge radius: **≥ 1 mm**

4. Sensor & Interlock System

The safety system relies on mandatory sensors and interlocks.

4.1 Mandatory Sensors

- Alignment sensor
- Lock engagement sensor
- Shroud presence sensor
- Over-temperature sensor
- Over-torque sensor
- Vibration anomaly sensor

4.2 Interlock Conditions

Operation must be prevented if:

- Alignment is incomplete
- Lock is not fully engaged
- Shroud is removed
- Temperature exceeds safe limit
- Torque exceeds safe limit
- Vibration signature indicates instability

4.3 Interlock Behavior

- Interlocks must be hardware-enforced
- Controller must log all interlock events
- Interlocks must prevent start and force stop

5. Emergency Stop Behavior

Emergency stop (E-Stop) is a universal, mandatory behavior.

5.1 E-Stop Trigger Conditions

- Manual E-Stop button
- Controller E-Stop command
- Any critical fault condition
- Any sensor indicating imminent hazard

5.2 E-Stop Response Requirements

- Torque must drop to zero within **20 ms**
- Shaft must decelerate to zero within safe stopping distance
- Lock must remain engaged
- Unlock must be disabled until safe
- Controller must log E-Stop event

5.3 Post-E-Stop State

- Module enters **Emergency Stop** state
- Only transitions allowed:
 - Emergency Stop → Fault
 - Fault → Idle (after manual reset)

6. Fault Handling & Logging

Fault handling ensures traceability and safe recovery.

6.1 Fault Categories

- **Critical Fault** — immediate hazard
- **Major Fault** — operational instability
- **Minor Fault** — non-hazardous anomaly

6.2 Required Fault Codes

- Over-temperature
- Over-torque
- Vibration anomaly
- Alignment failure
- Lock failure
- Shroud removal
- Electrical fault
- Controller communication fault

6.3 Logging Requirements

- Timestamp
- Fault category
- Fault code
- Sensor values at time of fault
- State machine state
- Recovery action taken

6.4 Recovery Behavior

- Minor faults: automatic recovery permitted
- Major faults: manual reset required
- Critical faults: E-Stop required

7. Safety Certification Requirements

A module is certified only if it:

- Implements all safety sensors
- Enforces all interlocks
- Meets all protective geometry requirements
- Passes all safety tests in the **Compliance Test Suite**
- Demonstrates safe E-Stop behavior
- Logs all required fault data
- Maintains hazard envelope integrity
- Meets timing guarantees for safety transitions

8. Safety Test Summary Table

Test	Objective	Pass Criteria
Shroud Integrity	Verify enclosure strength	No deformation, no breach
Interlock Enforcement	Validate interlock logic	All unsafe states blocked
E-Stop Response	Verify emergency behavior	Torque zero ≤ 20 ms
Fault Logging	Validate traceability	All fields logged correctly
Hazard Envelope	Verify spatial safety	No access to Zone 0/1